Part III will consist of a record of the invertebrate material taken in each haul, with a summary of the distribution of the invertebrate fauna of each section and of its seasonal and annual variation.

In Part IV endeavour will be made to summarise the economic and biological results of the whole work.

Explanation of Trawling Records.

The headings of results of individual hauls set forth in the first place the station number and serial prefix under which the haul is entered in our manuscript logs—as thus, "S. 10." The position of the haul is indicated by a section number in Roman numerals (as "Section XXII") corresponding to the Roman numerals on the two charts (Plates I and II) which accompany the Report. It will be observed that some hauls traverse parts of several sections.[1]

The position is further particularised by a statement of the distance and magnetic bearing of the station, at about its central part, from some landmark, and by soundings in fathoms. The nature of the bottom is denoted by the usual abbreviations, viz.:—

cl. = clay; crl. = coral; g. = gravel; m. = mud; oz. = ooze; r. = rock; s. = sand; sh. = shells; st. = stones; f. = fine; c. = coarse.

The various nets used are thus denoted:—B.T., beam trawl, of whatever length of beam is indicated, with mesh of the usual commercial size, 1½ inches between each knot of the cod end; Sh.T., shrimp trawl of about 9 feet beam; Sp.T., a large bag of sprat mesh fastened to the "back" of the beam trawl net between the mouths of the "pockets"; Mq.-T., a bag of mosquito muslin fastened to the "back" of the beam trawl net, just in front of the sprat mesh bag when both bags are used. The bags are so placed as to catch any small organism whirled upwards by the bight of the foot-rope through the meshes of the "back" of the trawl net. Being constant in size and position they probably give results comparable quantitatively between different hauls, but bearing an unascertainable relation to the quantity of small organisms which pass through the meshes of the whole net. Probably no net or combination of nets will give a reliable quantitative account of all the ground organisms large and small in the tract of the net; because, if the mesh of the main net be so far reduced as to retain small as well as large, it cannot be hauled fast enough to catch the more active of the larger creatures, and will therefore contain an undue proportion of the smaller. The device of enclosing the after portion of an ordinary trawl net in a larger net of fine mesh appears at first sight to be capable of yielding a quantitative

[1] A number of hauls lie outside the area covered by the sections, and are designated only by the heading, "Calf of Man." They are tabulated below for convenience of reference to East Coast hauls generally.

result of at least the fish, large and small, which enter the net and are severally retained by or escape through the ordinary commercial trawl mesh. In reality, however, the result is fallacious, because, as far as the writer can judge by experience, the presence of the outer bag alone impedes the efficiency of the whole net, while the extra rubbing pieces necessary to protect fine mesh still further lessen its capacity for capturing large fish. The same objection applies in a far less degree to bags attached only to the "back" of the trawl. They are relatively small and require no dead weight in the shape of rubbing pieces, and, though they catch a lot of small fish and invertebrates, they are high enough to avoid clogging with sand. The most, however, that one can claim for the quantitative significance of their record is that when they catch no fish smaller than those retained in the trawl net, the number of such smaller fish in the track of the trawl is insignificant, and vice versa. Probably this is as much as one needs to know of the grounds on which trawlers are accused of harrying, if not of actually catching, the "brood and nature of the fish."

The abbreviation N.D. means a naturalist's two-sworded dredge, 24 inches wide. Such a dredge is worked on a separate warp, and as a rule only for a short part of the time occupied by the trawl haul.

Hauls in which, by reason of accident to the net, the catch could not be considered as representative, are not enumerated below, while discussion of the invertebrate catches of all hauls are reserved for another part of the Report.

For the convenience of the general reader the marketable kinds are placed first in the lists of fishes, with the inevitable result that some unmarketable kinds are widely separated from their nearest relations. The contents of the trawl are set forth separately from those of the sprat mesh and mosquito muslin bags, and the record of each haul ends with a summary of the fishes of marketable kinds divided, in the case of the more important kinds into "unsaleable" and "saleable" and "immature" and "mature."

"Saleable" and "unsaleable" are arbitrary terms at best: for what will sell easily in one market or at seasons of scarcity may be accounted too small in another market or worthless when fish are plenty. In this report an "unsaleable" fish of whatever kind is one of less than 22 centimetres, about $8\frac{3}{4}$ inches.

"Mature" and "immature" are epithets which may legitimately be used in more than one sense, and in fishery legislative agitation "immature" seems to denote fish which are too small to secure the best market price per lb. or cwt. In this Report an "immature" fish is one of less size than that at which the female of the species appears usually to breed for the first time on the east coast of Ireland. While it is not claimed that the matter has been the subject of profound study in the course of the survey, the subjoined sizes of maturity will probably be exempt from accusation of exaggeration.

Species.	Female mature at cm.	Female mature at inches (about)	Species.	Female mature at cm.	Female mature at inches (about)
Plaice	32	12 9/32	Megrim	32	12 9/32
Sole	32	12 17/32	Cod	63	24 13/16
Turbot	46	18 1/8	Haddock	28	11 3/32
Brill	38	14 31/32	Whiting	25	9 27/32
Lemon Sole	25	9 27/32	Grey Gurnard	25	9 27/32
White Sole	32	12 9/32	Red Gurnard	25	9 27/32
Flounder	25	9 27/32	Latchet	32	12 9/32

Measurements of fish taken before 1st August, 1905, are given at the nearest centimetre of total length—thus a fish of less than 4·5 cm. is entered at 4 cm., and one of 4·5 cm. or more (up to 5·4 cm.) at 5 cm. After that date, on S. 298 and following stations, in order to harmonize the results with those of the International North Sea Investigations, the measurements were made in centimetres, the fractions being omitted—thus all fish of 4 cm and over, but less than 5 cm., are entered as 4 cm. In the case of skates and rays the measurement is the greatest width across the "wings."

Centimetres are to inches roughly as follows:—

cm.		in.	cm.		in.	cm.		in.
1	=	3/8	35	=	13 25/32	69	=	27 5/32
2	=	25/32	36	=	14 3/16	70	=	27 9/16
3	=	1 3/16	37	=	14 9/16	71	=	27 15/16
4	=	1 9/16	38	=	14 31/32	72	=	28 3/8
5	=	1 31/32	39	=	15 11/32	73	=	28 3/4
6	=	2 3/8	40	=	15 3/4	74	=	29 5/32
7	=	2 3/4	41	=	16 5/32	75	=	29 17/32
8	=	3 5/32	42	=	16 17/32	76	=	29 15/16
9	=	3 17/32	43	=	16 15/16	77	=	30 5/16
10	=	3 15/16	44	=	17 5/16	78	=	30 23/32
11	=	4 11/32	45	=	17 23/32	79	=	31 1/8
12	=	4 23/32	46	=	18 1/8	80	=	31 1/2
13	=	5 1/8	47	=	18 1/2	81	=	31 29/32
14	=	5 1/2	48	=	18 29/32	82	=	32 5/16
15	=	5 29/32	49	=	19 5/16	83	=	32 11/16
16	=	6 5/16	50	=	19 11/16	84	=	33 1/16
17	=	6 11/16	51	=	20 1/16	85	=	33 15/32
18	=	7 1/16	52	=	20 15/32	86	=	33 27/32
19	=	7 15/32	53	=	20 27/32	87	=	34 7/32
20	=	7 27/32	54	=	21 9/32	88	=	34 21/32
21	=	8 9/32	55	=	21 21/32	89	=	35 1/32
22	=	8 21/32	56	=	22 1/16	90	=	35 7/16
23	=	9 1/16	57	=	22 7/16	91	=	35 27/32
24	=	9 7/16	58	=	22 27/32	92	=	36 7/32
25	=	9 27/32	59	=	23 1/4	93	=	36 19/32
26	=	10 7/32	60	=	23 5/8	94	=	37 1/16
27	=	10 5/8	61	=	24 1/32	95	=	37 13/32
28	=	11 1/32	62	=	24 13/32	96	=	37 13/16
29	=	11 13/32	63	=	24 13/16	97	=	38 3/16
30	=	11 13/16	64	=	25 3/16	98	=	38 19/32
31	=	12 7/32	65	=	25 19/32	99	=	38 31/32
32	=	12 19/32	66	=	25 31/32	100	=	39 3/8
33	=	13	67	=	26 3/8			
34	=	13 3/8	68	=	26 25/32			

TRAWLING RECORDS.

S. 5.—30/11/'01. Section XII.

3-5½ mi. E. by N. of Balbriggan. 10-14 fath.
B.T. 40′. 7.20 a.m.—9.0 a.m. C.S.N.-T. M.S.N.-T. Mq.-T.
N.D. 8.0 a.m.—8.40 a.m.

IN TRAWL.

Plaice ..	cm.	18	19	20	21	23	28	30	31	32	33	36
	No.	*1*	*1*	*2*	*2*	*1*	*1*	*1*	*2*	*1*	*2*	*1*
	cm.	37										
	No.	*2*										
Dab ..	cm.	12	16	18	19	20	21	22	24	25	26	
	No.	*1*	*11*	*7*	*6*	*5*	*11*	*3*	*2*	*1*	*2*	
Brill ..	cm.	41										
	No.	*1*										
Grey gurnard	cm.	15	17	18	27	30	37	38				
	No.	*2*	*1*	*1*	*1*	*1*	*1*	*1*				
Red gurnard	cm.	14										
	No.	*2*										
Thornback ..	cm.	15	16	17	21	22	24	25	28	29		
	No.	*1*	*1*	*1*	*1*	*1*	*2*	*1*	*1*	*2*		
Spur dog ..	cm.	50	70									
	No.	*1*	*1*									
Dragonet ..	cm.	15	20									
	No.	*1*	*1*									

IN FINE-MESH NETS.

Goby ..	cm.	2·4	3·3
	No.	*1*	*1*
Solenette ..	cm.	7·5	
	No.	*1*	

SUMMARY OF MARKETABLE FISH IN TRAWL.

	Total.	Unsaleable.	Saleable.	Immature.	Mature.
Plaice ..	17	6	11	11	6
Dab ..	49	43	6	–	–
Grey gurnard	8	4	4	4	4
Red gurnard	2	2	0	2	0
Thornback ..	11	4	7	–	–

S. 12.—13/1/'02. Section XXIV.

1 mi. W.N.W. of Kish N. Buoy. 12-14 fath. s.sh.
B.T. 40′. 2.0 p.m.—4.0 p.m. Mq.-T. C.S.N.-T. M.S.N.-T.

IN TRAWL.

Plaice ..	cm.	19	20	21	22	23	24	25	26	27	28	29
	No.	*1*	*1*	*1*	*2*	*2*	*1*	*1*	*4*	*6*	*6*	*5*
	cm.	30	31	32	33	35	36	39	42			
	No.	*3*	*5*	*5*	*2*	*1*	*1*	*2*	*1*			
Dab ..	cm.	15	16	17	18	19	20	21	22	23	24	25
	No.	*4*	*4*	*17*	*17*	*6*	*11*	*13*	*5*	*6*	*11*	*4*
	cm.	26	28	29								
	No.	*5*	*2*	*2*								
Sole ..	cm.	24	25	26	34							
	No.	*2*	*1*	*1*	*1*							
Halibut ..	cm.	80										
	No.	*1*										
Cod	cm.	19	25	53								
	No.	*1*	*1*	*1*								
Whiting ..	cm.	16	18	21								
	No.	*1*	*1*	*1*								
Cuckoo ray	cm.	19	36									
	No.	*1*	*2*									
Spur dog ..	No.	*3*										

IN FINE-MESH NETS.

Freckled goby. Several.

SUMMARY OF MARKETABLE FISH IN TRAWL.

	Total.	Unsaleable.	Saleable.	Immature.	Mature.
Plaice ..	55	3	52	43	12
Dab ..	107	72	35	–	–
Sole ..	5	0	5	4	1
Halibut ..	1	0	1	–	–
Cod	3	1	2	3	0
Whiting ..	3	3	0	3	0
Cuckoo ray	3	1	2	–	–

S. 14.—14/1/'02. Section II.

3 mi. S.E. of Clogher Head. 11-12 fath.
B.T. 40′. 3.0 p.m.—4.0 p.m. Mq.-T. Sh. T. 4.25 p.m.—4.40 p.m.

IN TRAWL.

Plaice ..	cm.	20	25	26	27	28	29	30	31	32	33	34
	No.	*1*	*4*	*2*	*6*	*1*	*2*	*4*	*1*	*4*	*3*	*3*
	cm.	35	38									
	No.	*2*	*1*									

Flounder ..	cm.	34									
	No.	*1*									
Dab ..	cm.	13	15	16	17	18	20	21	25	26	
	No.	*1*	*1*	*2*	*5*	*4*	*2*	*1*	*1*	*1*	
Smearеen ..	cm.	20									
	No.	*1*									
Brill ..	cm.	29									
	No.	*1*									
Whiting ..	cm.	15	16	17	18	19	20	21	22	23	28
	No.	*4*	*3*	*1*	*3*	*6*	*4*	*1*	*1*	*1*	*1*
Grey gurnard	cm.	31									
	No.	*1*									
Herring ..	cm.	18									
	No.	*1*									

SUMMARY OF MARKETABLE FISH IN TRAWL.

	Total.	Unsaleable.	Saleable.	Immature.	Mature.
Plaice ..	34	1	33	21	13
Flounder ..	1	0	1	0	1
Dab ..	18	16	2	–	–
Smearеen ..	1	1	0	–	–
Brill ..	1	0	1	1	0
Whiting ..	25	22	3	24	1
Grey gurnard	1	0	1	0	1

S. 19.—16/1/'02. Section II.

1½ mi. off Clogher Head. 8–10 fath. f.s.
B.T. 34°. 9.55 a.m.—11.30 a.m. Mq.-T. N.D. 10.5 a.m.—10.55 a.m.

IN TRAWL.

Plaice ..	cm.	17	18	20	21	22	23	25	26	27	28	29
	No.	*3*	*2*	*2*	*1*	*3*	*3*	*3*	*3*	*1*	*4*	*2*
	cm.	30	32	33								
	No.	*1*	*1*	*1*								
Flounder ..	cm.	21										
	No.	*1*										
Dab ..	cm.	10	11	12	13	14	15	16	17	18	19	20
	No.	*1*	*8*	*9*	*5*	*16*	*13*	*8*	*9*	*10*	*2*	*5*
	cm.	21	22									
	No.	*1*	*2*									
White sole ..	cm.	36										
	No.	*1*										
Smearеen ..	cm.	20	21									
	No.	*1*	*1*									
Brill ..	cm.	28										
	No.	*1*										

Grey gurnard	cm.	8	12	13	15	16	19	35
	No.	*3*	*1*	*1*	*1*	*1*	*1*	*1*
Thornback ..	cm.	15	16	20	24			
	No.	*1*	*1*	*1*	*1*			
Dragonet ..	No.	*1*						

IN FINE MESH NETS.

Dab ..	cm.	4	5	6	7	8	9	10	11	12	14
	No.	*1*	*2*	*1*	*2*	*1*	*1*	*1*	*2*	*2*	*1*
Grey gurnard	cm.	7	8	9	11	13					
	No.	*2*	*1*	*2*	*1*	*1*					
Freckled goby	No.	*5*									
Gunnel ..	No.	*2*									
Scaldfish ..	cm.	3									
	No.	*1*									
Pipefish ..	No.	*1*									

SUMMARY OF MARKETABLE FISH IN TRAWL.

	Total.	Unsaleable.	Saleable.	Immature.	Mature.
Plaice ..	30	8	22	28	2
Flounder ..	1	1	0	1	0
Dab ..	89	87	2	–	–
Smeareen ..	2	2	0	–	–
Brill ..	1	0	1	1	0
White sole ..	1	0	1	0	1
Grey gurnard	9	8	1	8	1
Thornback ..	4	3	1	–	–

S. 20.—16/1/'02. Section III a.

3 mi. off Clogher Head. 12–13 fath. f.s. m.
B.T. 34′. 2.40 p.m.—4.10 p.m. Mq.-T. N.D. 3.22 p.m.—3.40 p.m.

IN TRAWL.

Plaice ..	cm.	18	20	23	25	26	28	29	34	37		
	No.	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*		
Dab ..	cm.	12	13	14	15	16	17	18	19	20	21	22
	No.	*3*	*3*	*5*	*17*	*31*	*29*	*13*	*12*	*4*	*3*	*3*
	cm.	23	24	30								
	No.	*1*	*1*	*1*								
Lemon sole	cm.	20										
	No.	*1*										
Smeareen ..	cm.	18	20	22								
	No.	*1*	*2*	*1*								

Whiting ..	cm.	20										
	No.	*1*										
Grey gurnard	cm.	12	15	16	17	19	20	21	23	29	30	32
	No.	*1*	*2*	*3*	*3*	*1*	*3*	*1*	*1*	*1*	*4*	*1*
	cm.	33	35	36	39							
	No.	*2*	*2*	*1*	*1*							
Dragonet ..	No.	2										
Rockling ..	cm.	25										
	No.	*1*										
Solenette ..	cm.	10										
	No.	*1*										

IN FINE-MESH NETS.

Dab ..	cm.	11	13	15
	No.	*1*	*2*	*1*
Smearcen ..	cm.	5	6	14
	No.	*1*	*2*	*1*

SUMMARY OF MARKETABLE FISH IN TRAWL.

	Total.	Unsaleable.	Saleable.	Immature.	Mature
Plaice ..	9	2	7	7	2
Dab ..	126	120	6	–	–
Lemon sole	1	1	0	1	0
Smearcen ..	4	3	1	–	–
Whiting ..	1	1	0	1	0
Grey gurnard	27	14	13	15	12

S. 21.—17/1/'02. Section II.

2 mi. off, Clogher to Dunany. 8–10 fath. m.s.
B.T. 34′. 8.30 a.m.—10.25 a.m. Mq.-T. N.D. 9.5 a.m.—9.30 a.m.

IN TRAWL.

Plaice ..	cm.	17	19	20	21	23	24	25	26	27	28	29
	No.	*1*	*1*	*1*	*1*	*2*	*2*	*3*	*6*	*2*	*6*	*3*
	cm.	30	31	32	33	34	38	40	50	52		
	No.	*3*	*2*	*1*	*4*	*2*	*1*	*1*	*1*	*1*		
Flounder ..	cm.	25	26	29	30	34	38					
	No.	*1*	*1*	*1*	*2*	*1*	*1*					
Dab ..	cm.	11	12	13	14	15	16	17	18	19	20	21
	No;	*2*	*7*	*7*	*5*	*12*	*12*	*15*	*11*	*8*	*7*	*4*
	cm.	22	23									
	No.	*2*	*1*									
White sole ..	cm.	27										
	No.	*1*										